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Class: T2 Subject: Agile Software Development and DevOps Lab

Experiment 3

Aim: To deploy multi service applications using Docker Compose

This experiment maps to LO4.
LO4: Containerize, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes, and cloud infrastructure tools.

Theory:

Docker Compose is a tool used to define and manage multiple Docker containers using a single YAML configuration file. In this experiment, a multi-service application is created using **Nginx as the web server** and **MySQL as the database service**. Docker Compose allows both containers to communicate over a common network and manages their creation, execution, and removal with simple commands. It also simplifies the deployment and management of containerized applications by allowing multiple services to be configured and started together from a single file.

Commands: **docker compose up**
Starts all the services defined in the compose.yml or docker-compose.yml file. If the containers don't exist, Docker creates them first. The logs are displayed in the terminal.
Example:
docker compose up

docker compose up -d
Starts all services in **detached mode**, which means they run in the background and your terminal remains available for other commands.
Example:
docker compose up -d

docker compose ps
Displays the status of all containers in the current Compose project, including whether they are running, stopped, and which ports are exposed.
Example:
docker compose ps

docker compose logs
Shows the log output from all services, making it useful for debugging and monitoring applications.
Example:
docker compose logs
To view logs for a specific service:
docker compose logs web

docker compose stop
Stops all running containers without removing them. Since the containers still exist, they can be started again later.
Example:
docker compose stop

docker compose start
Starts containers that were previously stopped using docker compose stop. It does not create new containers.
Example:
docker compose start

docker compose restart
Stops and then starts the containers again. This is useful after making configuration changes or if a service needs to be restarted.
Example:
docker compose restart

docker compose down

Stops and removes all containers, networks, and other resources created by Docker Compose. By default, named volumes are **not** removed, so your persistent data is kept.

Example:

docker compose down

docker compose down -v

Performs the same actions as docker compose down, but also removes **named volumes**. This deletes any persistent data stored in those volumes, such as databases.

Example:

docker compose down -v

docker compose images

Lists the Docker images used by the services in the current Compose project. This helps you verify which images and versions are being used.

Example:

docker compose images

`docker compose config - Validate Compose file

Steps to Run a Multi-Service Docker Compose Application

Step 1: Create a project directory

mkdir docker-compose-demo

cd docker-compose-demo

Step 2: Create a Docker Compose file

Create a file named **docker-compose.yml** with the following configuration. This example uses **three services**: a web server (Nginx), a database (MySQL), and a cache (Redis).

```
version: "3.9"

services:
  web:
    image: nginx:latest
    container_name: web-server
    ports:
      - "8080:80"
```

```
    depends_on:
      - db
      - cache

  db:
    image: mysql:8.0
    container_name: mysql-db
    environment:
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_DATABASE: demo_db
    ports:
      - "3306:3306"
    volumes:
      - mysql-data:/var/lib/mysql

  cache:
    image: redis:latest
    container_name: redis-cache
    ports:
      - "6379:6379"

volumes:
  mysql-data:
```

Step 3: Start all services

Run the following command to create and start all containers:

```
docker compose up
```

To run them in the background:

```
docker compose up -d
```

Step 4: Verify running services

Check the status of the containers:

```
docker compose ps
```

Step 5: View container logs

Display logs from all services:
docker compose logs
Or view logs for a specific service:
docker compose logs web

Step 6: Stop the services
Stop all running containers without removing them:
docker compose stop

Step 7: Start the stopped services
Restart the previously stopped containers:
docker compose start

Step 8: Restart all services
Restart the running containers:
docker compose restart

Step 9: View the images used
List the Docker images used by the Compose project:
docker compose images

Step 10: Stop and remove containers
Remove all containers and networks created by Docker Compose:
docker compose down

Step 11: Remove containers and volumes
To also delete the persistent MySQL data stored in the volume:
docker compose down -v

Services Used

- **Web Service:** Nginx (serves web content)
- **Database Service:** MySQL 8.0 (stores application data)
- **Cache Service:** Redis (provides in-memory caching)

Output Screenshots -

File Edit Selection View Go Run Terminal Help docker-compose.yml - Visual Studio Code [Administrator]

docker-compose.yml

C: > Users > 15L > Desktop > DockerComposeDemo > docker-compose.yml

```yaml
version: "3.8"

services:
  web:
    image: nginx:latest
    container_name: web_server
    ports:
      - "8080:80"

  database:
    image: mysql:8.0
    container_name: mysql_db
    environment:
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_DATABASE: sampledb
      MYSQL_USER: user
      MYSQL_PASSWORD: user123
    ports:
      - "3307:3306"
```

```
07/27/2026  03:00 PM               364 docker-compose.yml
               1 File(s)            364 bytes
               2 Dir(s)  474,740,912,128 bytes free

C:\Users\15L\Desktop\DockerComposeDemo>docker compose up -d
[+] Running 12/12
 ✔database 11 layers [⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿]      0B/0B      Pulled                                                                                                  17.2s
   ✔edf85873f64e Pull complete                                                                                                                           2.6s
   ✔6ef6c7b50a93 Pull complete                                                                                                                           0.7s
   ✔e3e5d1ac74c1 Pull complete                                                                                                                           0.8s
   ✔0d74d296605b Pull complete                                                                                                                           1.7s
   ✔297d04cfe470 Pull complete                                                                                                                           2.6s
   ✔4c8a3e0d4e4b Pull complete                                                                                                                           4.0s
   ✔a63160a5eda1 Pull complete                                                                                                                           5.1s
   ✔7534d1db9f8d Pull complete                                                                                                                           5.2s
   ✔409ec2dab01d9 Pull complete                                                                                                                          8.6s
   ✔ab24264a27e9 Pull complete                                                                                                                           4.1s
   ✔96d30d9fbee8 Pull complete                                                                                                                           4.7s
[+] Running 1/3
 - Network dockercomposedemo_default  Created                                                                                                            1.6s
 ✔Container web_server              Started                                                                                                              1.6s
 - Container mysql_db               Starting                                                                                                             1.6s
Error response from daemon: Ports are not available: exposing port TCP 0.0.0.0:3306 -> 0.0.0.0:0: listen tcp 0.0.0.0:3306: bind: Only one usage of each socket address (protocol/network address/port) is normall
y permitted.

C:\Users\15L\Desktop\DockerComposeDemo>docker compose down

C:\Users\15L\Desktop\DockerComposeDemo>docker compose down
[+] Running 3/3
 ✔Container web_server              Removed                                                                                                              0.5s
 ✔Container mysql_db                Removed                                                                                                              0.0s
 ✔Network dockercomposedemo_default  Removed                                                                                                             0.3s

C:\Users\15L\Desktop\DockerComposeDemo>docker compose up -d
[+] Running 2/3
 - Network dockercomposedemo_default  Created                                                                                                            0.6s
 ✔Container mysql_db                Started                                                                                                              0.6s
 ✔Container web_server              Started                                                                                                              0.4s

C:\Users\15L\Desktop\DockerComposeDemo>docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED              STATUS              PORTS                               NAMES
ea324727489d   mysql:8.0      "docker-entrypoint.s…"   About a minute ago   Up About a minute   33060/tcp, 0.0.0.0:3307->3306/tcp   mysql_db
a34b4efc6e58   nginx:latest   "/docker-entrypoint.…"   About a minute ago   Up About a minute   0.0.0.0:8080->80/tcp                web_server

C:\Users\15L\Desktop\DockerComposeDemo>
```

```
 - Network dockercomposedemo_default  Created
  ✔Container mysql_db                 Started
  ✔Container web_server               Started

C:\Users\15L\Desktop\DockerComposeDemo>docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED              STATUS              PORTS                               NAMES
ea32472749fd   mysql:8.0      "docker-entrypoint.s…"   About a minute ago   Up About a minute   33060/tcp, 0.0.0.0:3307->3306/tcp   mysql_db
a34b4efc6e58   nginx:latest   "/docker-entrypoint.…"   About a minute ago   Up About a minute   0.0.0.0:8080->80/tcp                web_server

C:\Users\15L\Desktop\DockerComposeDemo>docker compose ps
NAME         IMAGE          COMMAND                  SERVICE    CREATED         STATUS         PORTS
mysql_db     mysql:8.0      "docker-entrypoint.s…"   database   9 minutes ago   Up 9 minutes   33060/tcp, 0.0.0.0:3307->3306/tcp
web_server   nginx:latest   "/docker-entrypoint.…"   web        9 minutes ago   Up 9 minutes   0.0.0.0:8080->80/tcp

C:\Users\15L\Desktop\DockerComposeDemo>docker compose ps
NAME         IMAGE          COMMAND                  SERVICE    CREATED         STATUS         PORTS
mysql_db     mysql:8.0      "docker-entrypoint.s…"   database   9 minutes ago   Up 9 minutes   33060/tcp, 0.0.0.0:3307->3306/tcp
web_server   nginx:latest   "/docker-entrypoint.…"   web        9 minutes ago   Up 9 minutes   0.0.0.0:8080->80/tcp

C:\Users\15L\Desktop\DockerComposeDemo>docker compose stop
[+] Stopping 2/2
  ✔Container web_server  Stopped
  ✔Container mysql_db    Stopped

C:\Users\15L\Desktop\DockerComposeDemo>docker compose start
[+] Running 2/2
  ✔Container mysql_db    Started
  ✔Container web_server  Started

C:\Users\15L\Desktop\DockerComposeDemo>docker compose restart
[+] Restarting 2/2
  ✔Container web_server  Started
  ✔Container mysql_db    Started

C:\Users\15L\Desktop\DockerComposeDemo>docker compose images
CONTAINER           REPOSITORY          TAG                 IMAGE ID            SIZE
mysql_db            mysql               8.0                 6cd09145362d        799MB
web_server          nginx               latest              4e5db4761e0f        161MB

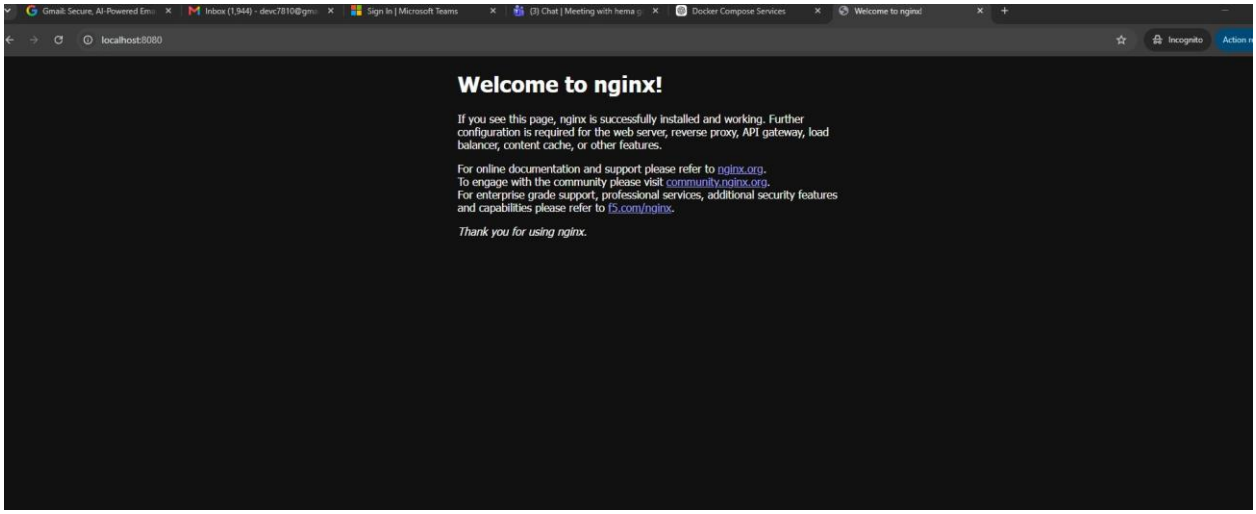
C:\Users\15L\Desktop\DockerComposeDemo>docker compose config
name: dockercomposedemo
services:
  database:
    container_name: mysql_db
    environment:
      MYSQL_DATABASE: sampledb
      MYSQL_PASSWORD: user123
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_USER: user
    image: mysql:8.0
    networks:
```

```
C:\Users\15L\Desktop\DockerComposeDemo>docker compose images
CONTAINER           REPOSITORY          TAG                 IMAGE ID            SIZE
mysql_db            mysql               8.0                 6cd09145362d        799MB
web_server          nginx               latest              4e5db4761e0f        161MB

C:\Users\15L\Desktop\DockerComposeDemo>docker compose config
name: dockercomposedemo
services:
  database:
    container_name: mysql_db
    environment:
      MYSQL_DATABASE: sampledb
      MYSQL_PASSWORD: user123
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_USER: user
    image: mysql:8.0
    networks:
      default: null
    ports:
      - mode: ingress
        target: 3306
        published: "3307"
        protocol: tcp
  web:
    container_name: web_server
    image: nginx:latest
    networks:
      default: null
    ports:
      - mode: ingress
        target: 80
        published: "8080"
        protocol: tcp
networks:
  default:
    name: dockercomposedemo_default

C:\Users\15L\Desktop\DockerComposeDemo>docker compose down
[+] Running 3/3
  ✔Container mysql_db                 Removed
  ✔Container web_server               Removed
  ✔Network dockercomposedemo_default  Removed

C:\Users\15L\Desktop\DockerComposeDemo>
```


Gmail: Secure, AI-Powered Ema
Inbox (1,944) - devc7810@gma
Sign In | Microsoft Teams
(3) Chat | Meeting with hema g
Docker Compose Services
Welcome to nginx!
localhost:8080
Incognito
Welcome to nginx!
If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.
For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.
Thank you for using nginx.